

Solutions to Problem Set 1

1. Let a discrete random variable X takes values in the set $\{x_1, x_2, \dots, x_N\}$. Prove that the entropy of X satisfies the inequality

$$H(X) \leq \log N$$

and with equality if and only if the probability $p_i = 1/N$ for all i . (Hints: review the slides)

Solution. Let $\{p_1, p_2, \dots, p_N\}$, $\{q_1, q_2, \dots, q_N\}$ be two probability distributions on the set. For the relative entropy,

$$\begin{aligned} D(p||q) &= \sum_{i=1}^N p_i \log_2 \left(\frac{p_i}{q_i} \right) \\ &= - \sum_{i=1}^N p_i \log_2 \left(\frac{q_i}{p_i} \right) \\ &\geq - \sum_{i=1}^N p_i \left(\frac{q_i}{p_i} - 1 \right) = 0 \end{aligned}$$

Therefore, $D(p||q) \geq 0$ with the equality holds if and only if p and q have the same distribution.

Let q be the uniform distribution, i.e. $q_i = 1/N$ for all i . For any distribution p ,

$$\begin{aligned} D(p||q) &= \sum_{i=1}^N p_i \log_2 p_i + \log_2 K \sum_{i=1}^N p_i \\ &= -H(X) + \log_2 K \end{aligned}$$

Since $D(p||q) \geq 0$, we conclude that $H(X) \leq \log N$ with equality if and only if the probability $p_i = q_i = 1/N$ for all i .

2. A discrete memoryless source has an alphabet of 7 symbols whose probabilities of occurrence are as described here:

$$\begin{pmatrix} s_0 & s_1 & s_2 & s_3 & s_4 & s_5 & s_6 \\ 0.25 & 0.25 & 0.125 & 0.125 & 0.125 & 0.0625 & 0.0625 \end{pmatrix}$$

- Compute the Huffman code for the source. When combining, give a priority to the more 'combined' symbol. Compute the mean and variance of the codeword length.
- Compute the Huffman code for the source. When combining, give a priority to the less 'combined' symbol. Compute the mean and variance of the codeword length.
- Verify the entropy for the source equals to the average codeword length in a) and b). Explain why the efficiency is 100%.
- Compare the variance of the codeword length in a) and b) and comment on the result.

Solution. For a) and b), the constructed Huffman tree should reflect the priority.

The codeword length of s_0, s_1 is 2, length of s_2, s_3, s_4 is 3, and length of s_5, s_6 is 4. Therefore, the average codeword length and variance are

$$\bar{L} = (0.25 + 0.25) \times 2 + (0.125 + 0.125 + 0.125) \times 3 + (0.0625 + 0.0625) \times 4 = \frac{21}{8} = 2.625$$

$$\begin{aligned} \text{Var}(L) &= (0.25 + 0.25) \times 0.625 + (0.125 + 0.125 + 0.125) \times 0.375 + (0.0625 + 0.0625) \times 1.375 \\ &= \frac{31}{64} \approx 0.484 \end{aligned}$$

- c) The entropy of the source is

$$\begin{aligned} H(X) &= -[(0.25 + 0.25) \times \log_2(0.25) + (0.125 + 0.125 + 0.125) \times \log_2(0.125) \\ &\quad + (0.0625 + 0.0625) \times \log_2(0.0625)] \\ &= \frac{21}{8} = 2.625 = \bar{L} \end{aligned}$$

The reason for achieving 100% efficiency is that the probabilities can be expressed as $p_i = 2^{-l_i}, l_i \in \mathbb{N}$. Therefore, one can construct a Huffman code, assigning a codeword of length l_i to s_i , that satisfies Kraft's inequality and

$$\bar{L} = \sum_{i=0}^{N-1} p_i l_i = \sum_{i=0}^{N-1} p_i (-\log_2(p_i)) = H(X)$$

- d) Variance is same in this case because the lengths of the codewords are the same.

(Question: Is this conclusion universal?)

3. Prove the chain rule of entropy

$$H(X_1 X_2 \cdots X_k) = \sum_{i=1}^k H(X_i | X_{i-1} \cdots X_1).$$

Solution. We prove by induction. For $k = 1$, $H(X_1) = H(X_1)$. Suppose it is true for $k - 1$, i.e.

$$H(X_1 X_2 \cdots X_{k-1}) = \sum_{i=1}^{k-1} H(X_i | X_{i-1} \cdots X_1).$$

For case k , if we see $X_1 X_2 \cdots X_{k-1}$ as one random variable, then

$$\begin{aligned} H(X_1 X_2 \cdots X_k) &= H(X_1 X_2 \cdots X_{k-1}) + H(X_k | X_{k-1} \cdots X_1) \\ &= \sum_{i=1}^{k-1} H(X_i | X_{i-1} \cdots X_1) + H(X_k | X_{k-1} \cdots X_1) \\ &= \sum_{i=1}^k H(X_i | X_{i-1} \cdots X_1). \end{aligned}$$

4. For random variables of given distributions

$$X \sim \begin{pmatrix} -2 & -1 & 0 & 1 & 2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \end{pmatrix}, \quad Y = X^2,$$

find $H(XY)$, $H(Y|X)$, $H(X|Y)$, $I(X;Y)$.

Solution. Since Y is a function of X ,

$$H(XY) = H(X) = \log_2(5) = 2.32, \quad H(Y|X) = H(XY) - H(X) = 0$$

Then we calculate the entropy of Y , since

$$Y \sim \begin{pmatrix} 0 & 1 & 2 \\ 0.2 & 0.4 & 0.4 \end{pmatrix}$$

$$H(Y) = -[0.2 \log_2(0.2) + (0.4 + 0.4) \log_2(0.4)] = 1.52$$

$$H(X|Y) = H(XY) - H(Y) = 0.80, \quad I(X|Y) = H(Y) - H(Y|X) = 1.52$$

5. Prove the following properties of differential entropy:

a) For a random variable X and $a \neq 0$,

$$h(aX) = h(X) + \log |a|.$$

b) For two distributions $p(x), q(x)$ and a random variable X that follows $p(x)$,

$$h(X) \leq - \int p(x) \log q(x) dx$$

and with equality if and only if $p(x) = q(x)$.

c) For a random variable X ,

$$h(X) \leq \frac{1}{2} \log(2\pi e \sigma^2), \quad \text{if } \mathbb{E}(X^2) \leq \sigma^2$$

and with equality if and only if X is Gaussian. (Hints: use b))

Solution. a) Let $Y = aX$, we have $p_Y(y) = p_X(x)/|a|$, then

$$\begin{aligned} h(aX) &= h(Y) = \int -p_Y(y) \log p_Y(y) dy \\ &= \int -(p_X(x)/|a|) \log(p_X(x)/|a|) |a| dx \\ &= - \int p_X(x) \log p_X(x) dx + \log |a| \\ &= h(X) + \log |a| \end{aligned}$$

b) We subtract right-hand-side from left-hand-side,

$$\begin{aligned} h(X) + \int p(x) \log q(x) dx &= \int p(x) \log \frac{q(x)}{p(x)} dx \\ &\leq \int p(x) \left(\frac{q(x)}{p(x)} - 1 \right) dx \\ &= 0 \end{aligned}$$

with equality if and only if the inequality $\log x \leq x - 1$ takes equality, when $p(x) = q(x)$.

c) Let $X \sim p(x)$, and $q(x) \sim \mathcal{N}(0, \sigma^2)$, from b) we have:

$$\begin{aligned} h(X) &\leq - \int p(x) \log \left(\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}} \right) dx \\ &= \int p(x) \left(\frac{x^2}{2\sigma^2} + \log(\sqrt{2\pi}\sigma) \right) dx \\ &= \frac{1}{2\sigma^2} \int x^2 p(x) dx + \frac{1}{2} \log(2\pi\sigma^2) \\ &\leq \frac{1}{2} \log(2\pi e \sigma^2) \end{aligned}$$

with equality if and only if $p(x) = q(x)$, i.e. X is Gaussian with variance σ^2 .

6. (Challenge) We will investigate Gaussian channels in this question. We define the entropy of random vector $\vec{X} = \{X_1, X_2, \dots, X_n\}$ as the joint entropy of the random variables in the vector, i.e., $H(\vec{X}) = H(X_1 X_2 \dots X_n)$.

- a) Consider a Gaussian random vector $\vec{X} \sim \mathcal{N}(\vec{\mu}, \mathbf{K}_{\vec{X}})$, show that

$$h(\vec{X}) = \frac{1}{2} \log [(2\pi e)^n \det(\mathbf{K}_{\vec{X}})] .$$

- b) Assume the input is $\vec{X} \sim \mathcal{N}(\vec{\mu}, \mathbf{K}_{\vec{X}})$, the channel matrix is $\mathbf{H}$, and the additive Gaussian noise is $\vec{Z} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$. Then the output is $\vec{Y} = \mathbf{H}\vec{X} + \vec{Z}$, show that

$$I(\vec{X}; \vec{Y}) = \frac{1}{2} \log \det \left(\mathbf{I} + \frac{\mathbf{H} \mathbf{K}_{\vec{X}} \mathbf{H}^T}{\sigma^2} \right) .$$

- c) Consider a parallel Gaussian channel

$$\begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} + \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} ,$$

where

$$\begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} \sim \mathcal{N} \left(0, \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \right) .$$

With power constraint $\mathbb{E}(X_1^2) + \mathbb{E}(X_2^2) \leq P$, find the capacity

$$C = \max_{p_{X_1 X_2}(x_1, x_2)} I(\vec{X}; \vec{Y}) .$$

Solution. a) The density of a Gaussian random vector $\vec{X} \sim \mathcal{N}(\vec{\mu}, \mathbf{K}_{\vec{X}})$ is:

$$p(\vec{X}) = p(x_1, \dots, x_n) = \frac{1}{\sqrt{(2\pi)^n |\mathbf{K}_{\vec{X}}|}} e^{-\frac{1}{2}(\vec{X} - \vec{\mu})^T \mathbf{K}_{\vec{X}}^{-1} (\vec{X} - \vec{\mu})} \quad (1)$$

Thus the entropy of $\vec{X}$ is

$$\begin{aligned} h(\vec{X}) &= - \int p(\vec{X}) \log p(\vec{X}) \, d\vec{X} \\ &= - \int p(\vec{X}) \log \left[\frac{1}{\sqrt{(2\pi)^n |\mathbf{K}_{\vec{X}}|}} e^{-\frac{1}{2}(\vec{X} - \vec{\mu})^T \mathbf{K}_{\vec{X}}^{-1} (\vec{X} - \vec{\mu})} \right] \, d\vec{X} \\ &= \log \left[(2\pi)^{\frac{n}{2}} |\mathbf{K}_{\vec{X}}^{-1}|^{\frac{1}{2}} \right] + \frac{1}{2} \int p(\vec{X}) (\vec{X} - \vec{\mu})^T \mathbf{K}_{\vec{X}}^{-1} (\vec{X} - \vec{\mu}) \, d\vec{X} \end{aligned}$$

Let $\vec{Y} = \mathbf{K}_{\vec{X}}^{-\frac{1}{2}}(\vec{X} - \vec{\mu})$, then $\vec{Y} \sim \mathcal{N}(0, \mathbf{I}_n)$. We have $p(\vec{X}) = |\mathbf{K}_{\vec{X}}|^{-\frac{1}{2}}p(\vec{Y})$ and $d\vec{X} = |\mathbf{K}_{\vec{X}}|^{\frac{1}{2}} d\vec{Y}$. Substituting to have

$$\begin{aligned} h(\vec{X}) &= \log \left[(2\pi)^{\frac{n}{2}} |\mathbf{K}_{\vec{X}}|^{\frac{1}{2}} \right] + \frac{1}{2} \int p(\vec{Y}) \vec{Y}^T \vec{Y} d\vec{Y} \\ &= \frac{1}{2} \log \left[(2\pi)^n \det(\mathbf{K}_{\vec{X}}) \right] + \frac{1}{2} \sum_{i=1}^n \mathbb{E}[y_i^2] \\ &= \frac{1}{2} \log \left[(2\pi e)^n \det(\mathbf{K}_{\vec{X}}) \right]. \end{aligned}$$

b) We directly calculate the mutual information:

$$\begin{aligned} I(\vec{X}; \vec{Y}) &= I(\vec{X}; \mathbf{H}\vec{X} + \vec{Z}) = h(\mathbf{H}\vec{X} + \vec{Z}) - h(\mathbf{H}\vec{X} + \vec{Z}|\vec{X}) \\ &= h(\mathbf{H}\vec{X} + \vec{Z}) - h(\vec{Z}) \end{aligned}$$

By the linear transformation of Gaussian random vector, $Y \sim \mathcal{N}(\mathbf{H}\vec{\mu}, \mathbf{H}\mathbf{K}_{\vec{X}}\mathbf{H}^T + \sigma^2\mathbf{I}_n)$, thus

$$\begin{aligned} I(\vec{X}; \vec{Y}) &= \frac{1}{2} \log \left[(2\pi e)^n \det(\mathbf{H}\mathbf{K}_{\vec{X}}\mathbf{H}^T + \sigma^2\mathbf{I}_n) \right] - \frac{1}{2} \log \left[(2\pi e)^n \det(\sigma^2\mathbf{I}_n) \right] \\ &= \frac{1}{2} \log \det \left(\mathbf{I}_n + \frac{\mathbf{H}\mathbf{K}_{\vec{X}}\mathbf{H}^T}{\sigma^2} \right). \end{aligned}$$

c) Using the result from b), let $\mathbf{H} = \mathbf{I}_2$, we have $I(\vec{X}; \vec{Y}) = h(\vec{X} + \vec{Z}) - h(\vec{Z})$. To maximize mutual information,

$$\begin{aligned} C &= \max_{p_{X_1, X_2}(x_1, x_2)} \left\{ h(\vec{Y}) - h(\vec{Z}) \right\} \\ &= \max_{p_{X_1, X_2}(x_1, x_2)} \left\{ h(\vec{Y}) \right\} - \sum_{i=1}^2 \frac{1}{2} \log(2\pi e \sigma_i^2) \\ &\leq \sum_{i=1}^2 \left\{ \frac{1}{2} \log(2\pi e (P_i + \sigma_i^2)) - \frac{1}{2} \log(2\pi e \sigma_i^2) \right\} \\ &= \frac{1}{2} \log \left(1 + \frac{P_1}{\sigma_1^2} \right) + \frac{1}{2} \log \left(1 + \frac{P_2}{\sigma_2^2} \right) \\ &= \frac{1}{2} \log \left(\frac{(P_1 + \sigma_1^2)(P_2 + \sigma_2^2)}{\sigma_1^2 \sigma_2^2} \right), \end{aligned}$$

where the equality is achieved if and only if X_1, X_2 are independent Gaussian random variables.

Finally, because

$$(P_1 + \sigma_1^2)(P_2 + \sigma_2^2) \leq \left(\frac{P_1 + P_2 + \sigma_1^2 + \sigma_2^2}{2} \right)^2 = \left(\frac{P + \sigma_1^2 + \sigma_2^2}{2} \right)^2,$$

the capacity $C = \log \left(\frac{P + \sigma_1^2 + \sigma_2^2}{2\sigma_1\sigma_2} \right)$ is achieved when $P_1 + \sigma_1^2 = P_2 + \sigma_2^2$.